

World-Model Formal Concept Analysis: Exact Recovery, Provenance-Safe Revision, and Contextual Envelopes

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Abstract

Formal Concept Analysis (FCA) extracts concepts, implications, and a concept lattice from a binary object–attribute relation. Real knowledge stores are usually less obliging: an incidence can be false, unobserved, explicitly indeterminate, duplicated by a pipeline, revised, or later forgotten. We present a world-model (WM) formulation in which the primary object is a finite state of provenance-stamped observations and classical FCA is an exact late projection. The implementation runs natively in CeTTa and supports immutable revision, forgetting, cell-level explanations, source/dependence-group gates, bounded concept and implication enumeration, and robust/permissive incidence envelopes. A Lean 4 development proves exact order-isomorphic recovery of classical FCA, duplicate-idempotent revision, disjoint revision/forgetting laws, dependence-group invariance, and contextual-envelope obligations without **sorry**, wishlist stubs, or new axioms.

We first establish exact parity on 28 finite contexts, covering published examples, a 13-context Mizar family, and FCAPy controlled random contexts. The checked artifacts comprise 3,275 concepts, 1,330 canonical implications, and 11,734 cover edges. A further 19 wider contexts, including the 403×67 Bob Ross matrix, supply 1,000 exact extent/closure queries without unbounded lattice enumeration. We then run a frozen 405-case corruption study. Dependence-aware WM projections are invariant to correlated copies in every case, while raw/source counting is invariant in only a 0.007 fraction of cases. A permissive WM point projection improves the descriptive mean recovery of cells, concepts, implications, and covers over the crisp single-source projection; a robust projection does worse, showing that the WM degrees of freedom are substantive rather than uniformly beneficial. The robust/permissive envelope raises cell and closure coverage at the explicit cost of nonzero interval width; its midpoint also has lower Brier score and ECE than the crisp hard forecast in every frozen corruption stratum. This scores the midpoint as a point forecast; it does not establish calibration of the set-valued interval. In a separately frozen incomplete-context comparison, the permissive WM projection achieves better descriptive-mean point and structural recovery than both the lower and upper completions. The completion interval, however, has higher cell coverage and lower midpoint Brier score and ECE than the narrower WM envelope. Thus the experiment exposes a coverage–width tradeoff rather than a uniform winner. The study is descriptive, not an inferential significance claim, and uses synthetic corruption rather than a natural-domain uncertainty corpus.

1 Introduction

Formal Concept Analysis begins with a formal context $K = (G, M, I)$: objects G , attributes M , and a binary incidence relation $I \subseteq G \times M$ [1, 2]. From the Galois connection

$$A' = \{m \in M : \forall g \in A, gIm\}, \quad B' = \{g \in G : \forall m \in B, gIm\},$$

it derives concepts (A, B) satisfying $A' = B$ and $B' = A$, attribute implications, and a complete concept lattice. This is an exact and useful theory once I is fixed. The difficult step in an evidential system is often the condition *once I is fixed*.

A missing database cell is not a negative observation. Two reports can be nominally distinct but descend from the same experiment. An observation can be retracted without invalidating unrelated evidence. Different admissible extraction policies can induce different crisp relations, and a query may need the range across those policies rather than one prematurely selected matrix. Encoding all of these cases directly as Boolean incidence loses information before concept formation begins.

We therefore put a small world-model calculus in front of classical FCA. States, revision, forgetting, query, and explanation are primary. A gate projects one evidence profile into a crisp relation only when an exact FCA artifact is requested. A family of admissible models yields lower (robust) and upper (permissive) relations. The formulation is deliberately not restricted to PLN truth values: counts, four-valued observations, interval evidence, or other ordered evidence carriers can instantiate the interface.

The contributions are:

1. a finite provenance-state semantics with immutable revision and forgetting, distinct false, unknown, and missing statuses, and dependence-aware extraction;
2. native CeTTa operations for exact FCA projection, bounded NextClosure-style enumeration, canonical implications, cover edges, packed scale contexts, and cell-level provenance explanations;
3. Lean proofs connecting the state semantics to classical FCA and establishing revision, forgetting, deduplication, dependence, and contextual-envelope laws;
4. exact parity against independent FCA oracles on 28 enumeration contexts and 19 wider query contexts; and
5. a frozen corruption study that separates point-recovery quality, interval coverage/width, and invariance to duplicated evidence, together with a separately frozen lower/upper-completion FCA comparison.

The central result is not that one WM gate is always superior. It is that classical FCA is recovered exactly when the state is complete and the model family is a singleton, while richer state and model structure can be retained and tested when those assumptions do not hold.

2 From observations to formal contexts

2.1 Provenance states

Let S be a finite type of stamps. A provenance catalog fixes, for every $s \in S$, its source, declared dependence group, and query-local evidence contribution:

$$C_{\text{src}} : S \rightarrow \text{Source}, \quad C_{\text{grp}} : S \rightarrow \text{Group}, \quad C_{\text{obs}} : S \times G \times M \rightarrow E.$$

A stamp denotes immutable content. Reusing a stamp with conflicting source, group, or observation is therefore unrepresentable in the formal model and rejected by the runtime.

A state $W \in \mathcal{P}_{\text{fin}}(S)$ is the set of active stamps. For an additive evidence carrier E ,

$$e_W(g, m) = \sum_{s \in W} C_{\text{obs}}(s, g, m).$$

Revision and forgetting are

$$\text{Revise}(W, \Delta) = W \cup \Delta, \quad \text{Forget}(D, W) = W \setminus D.$$

Union makes exact replay idempotent. Additivity is asserted only for disjoint stamp sets:

$$W \cap \Delta = \emptyset \implies e_{W \cup \Delta} = e_W + e_\Delta.$$

This guard is important. Treating idempotent union as an additive monoid would silently count a replayed observation twice.

The executable carrier specializes observations to four statuses: $\{\mathbf{T}, \mathbf{F}, \mathbf{U}, \mathbf{M}\}$. $\mathbf{F}$ is an observed negative; $\mathbf{U}$ is an explicit indeterminate report; $\mathbf{M}$ means the state contains no applicable observation. Explanations return the contributing stamps with source and group metadata rather than only the projected bit.

2.2 Gates and exact FCA projection

An evidence gate $q : E \rightarrow \{\text{false}, \text{true}\}$ induces

$$I_{W,q}(g, m) \iff q(e_W(g, m)).$$

The implementation includes exact four-status extraction and thresholds over positive observations, distinct sources, and distinct dependence groups. Classical FCA is then applied to $I_{W,q}$, not redefined. In particular, the Lean map `formedConceptOrderIsoCrispConcept` is an order isomorphism between exact formed concepts and the standard FCA concept carrier for the same gated relation. Thus the WM lane is a conservative extension, not a competing notion of crisp concept.

Dependence groups alter what is counted. Let

$$\Gamma_W(g, m) = \{C_{\text{grp}}(s) : s \in W, C_{\text{obs}}(s, g, m) \neq 0\}.$$

The group-count statistic is $|\Gamma_W(g, m)|$. If every group contributed by Δ is already represented in W at every cell, then

$$|\Gamma_{W \cup \Delta}(g, m)| = |\Gamma_W(g, m)|$$

and every FCA family extracted through a group-count threshold is unchanged. This is a semantic guarantee about a declared dependence partition; it does not prove that the declaration itself is scientifically correct.

2.3 Contextual envelopes

A contextual model $\theta = (\mu_\theta, q_\theta)$ contains an evidence profile and a gate. For a nonempty admissible family Θ , define

$$\begin{aligned} \underline{I}(g, m) &\iff \forall \theta \in \Theta, q_\theta(\mu_\theta(g, m)), \\ \bar{I}(g, m) &\iff \exists \theta \in \Theta, q_\theta(\mu_\theta(g, m)). \end{aligned}$$

Then $\underline{I} \subseteq \bar{I}$. We call the pair an incidence envelope. It supports robust and permissive FCA queries without treating an unobserved cell as false. For $\Theta = \{\theta\}$, both relations reduce exactly to I_θ , and both concept-family projections reduce to ordinary FCA on that relation.

The envelope is intentionally a relation-level construction. The concept family of $\underline{I}$ is not generally the set-theoretic intersection of the concept families of all θ , and similarly for $\bar{I}$. Making the aggregation point explicit prevents an invalid interchange between Galois closure and model-family quantification.

3 Implementation and formal assurance

3.1 Native CeTTa execution

The public implementation is a CeTTa library consumed by a benchmark suite. Small contexts use transparent MeTTa definitions as differential specifications. Scale contexts use validated packed binary rows and native operations for:

- row and event-state construction;
- status, evidence, source, group, and explanation queries;
- immutable revise/remember/forget operations;
- column-index construction and batched extent/closure queries;
- replayable closure, implication, and concept certificates, retaining the indexed state, gate, query, intermediate extent, and closure, and exposing the exact finite boundary of consulted cells; and
- closed-intent enumeration, Duquenne–Guigues canonical bases, and Hasse cover edges under explicit output/step bounds.

The bounds are part of the semantic contract: exhausting a bound fails loudly rather than returning a prefix as a complete lattice. Contexts with at most 10 attributes are assigned to exact enumeration in the scale protocol. Wider contexts are query-only. This is not a claim that a general MeTTa runtime should outperform specialized FCA software.

An earlier recursive state/query path was replaced by bounded native carriers. No benchmark in this paper uses the retired expansion of a large packed matrix into one interpreted term per cell. Malformed rows, invalid statuses, out-of-universe observations, conflicting stamp reuse, and exceeded bounds have negative tests in addition to ordinary positive examples.

3.2 Lean obligations

The new Lean module builds on the existing FCA recovery bridge. Its primary definitions are a finite stamp set, immutable catalog, union revision, set difference forgetting, group support, exact concept-family extraction, and a contextual envelope. The audited theorems include:

- `revise_self`, `revise_comm`, and `revise_assoc`;
- `forget_revision_of_disjoint` and `evidence_forget_revision_of_disjoint`;
- `evidence_revise_of_disjoint`, exposing rather than hiding the freshness condition;
- `groupCountConceptFamily_revise_eq_left_of_groups_subset`;
- `finiteConceptFamily_forget_revision_of_disjoint`;
- the robust-to-permissive implication, singleton reductions, and contextual-envelope validity; and
- positive and negative executable canaries distinguishing a correlated stamp, an independent group, and an exact duplicate.

Table 1: Exact-enumeration validation. Artifact columns are totals within a family; every concept, canonical implication, and cover edge is compared as a set, not only by count.

Family	Contexts	Object $\times$ attribute range	Concepts	Basis	Covers
Published FCA examples	6	8–35 $\times$ 7–14	344	104	887
Mizar family	13	27–143 $\times$ 1–32	767	986	1,729
FCAPy controlled random	9	10–100 $\times$ 10	2,164	240	9,118

Table 2: Query-only scale validation. Maximum size is the largest context in the family; no complete lattice is claimed for these cases.

Family	Contexts	Maximum size	Exact queries
FCAPy controlled random	18	100 $\times$ 50	918
Bob Ross	1	403 $\times$ 67	82

The module contains no `sorry`, `theorem_wanted`, or declared axiom. Kernel dependency inspection of the load-bearing theorems reports only Lean’s standard `propext`, `Classical.choice`, and `Quot.sound` where needed. A native-evaluation canary was deliberately replaced by an ordinary kernel proof, so no native-decision trust axiom remains in the audited claims. These theorems formalize the abstract state and projection laws; they are not a formal verification of the C runtime.

4 Exact validation

4.1 Datasets and independent oracles

The exact suite contains three families:

1. six published FCA examples: Planets, Living Beings, Animals, Binary Relations, Tea Lady, and Music;
2. 13 contexts extracted from a pinned Mizar family; and
3. nine 10-attribute controlled-random contexts reproduced from the FCAPy benchmark generator [8].

Planets is checked against the published `fcaR` concepts and implications [7]. The remaining exact contexts are reconstructed by a lectic `NextClosure` implementation, an independently ordered level-wise implementation, and the `concepts` FCA package; lattice neighbors are also compared. Inputs and dependency versions are content-addressed.

The 19 wider contexts, including the FiveThirtyEight Bob Ross matrix [9], are not silently omitted. They have a separately frozen query protocol containing the empty and full attribute sets, every singleton, fixed prefixes, and fixed adjacent pairs. Direct set-intersection oracles and `concepts` agree on all extents and closures.

4.2 Revision and forgetting

The event protocol contains 14 operations: forgetting and restoring a packed stamp, repeating the same stamp, adding and removing a conflict, adding and removing explicit unknown evidence, corroboration, a multi-observation revision, and scoped forgetting by group, source, context, and

cell. Both a 30×30 random context and Bob Ross are checked at the initial state and after every event, yielding 30 prefix states. At each prefix,

$$Q(\text{foldUpdates}(W_0, e_1, \dots, e_t)) = Q(\text{buildFresh}(\text{survivors}_t))$$

for the frozen status, evidence, provenance, extent, and closure queries. Independent direct queries also agree with **concepts**. This validates incremental semantics against fresh recomputation; it does not prove an asymptotic update advantage.

4.3 Staged ergonomics measurement

A small receipt-backed campaign measures only fixtures whose golden outputs were already validated. Each case has one warm-up and five measured process invocations; an invocation is retained only when its output is byte-identical to the expected output. Timings include process startup, parsing, native execution, and golden validation. They are therefore end-to-end observations on one machine, not kernel-only timings or comparisons with specialized FCA engines.

Table 3: Descriptive native-runtime ergonomics. Rates have different work units by row and should not be compared across rows.

Case	Work per invocation	Median ms	Units/s	Golden
Bob Ross query batch	82 closure queries	9.989	8,209	yes
Immutable event replay	14 event operations	12.868	1,088	yes
Certificate checks	24 verified assertions	13.266	1,809	yes

The recorded campaign peak child RSS is 19.5 MiB. The public WM/FCA library has 2,309 lines (79,938 bytes), 110 type declarations, and 249 equations. The certificate gate covers 14 named positive, negative, tamper, cell boundary, history, and typed-wrapper behaviors. These counts measure source and test surface, not semantic complexity. The complete environment, input, binary, runner, protocol, program, expected-output, and library hashes are in receipt `6fdc93a90a89...`

5 Frozen corruption study

5.1 Protocol

The protocol was frozen before the complete run and is identified by `7dcd2936b014...` It crosses:

- 9 FCApy contexts with 10 attributes and 10, 30, or 100 objects at densities 0.1, 0.5, and 0.9;
- 5 deterministic seeds;
- missingness rates $\{0.10, 0.30, 0.50\}$;
- bit-flip rates $\{0.00, 0.05, 0.10\}$; and
- an explicit unknown rate of 0.05.

The Cartesian product contains 405 cases. Each case has a primary layer, an exact matrix copy from a distinct nominal source in the same dependence group, and an independently corrupted layer in another group.

Six models were fixed:

Table 4: Descriptive means across all 405 frozen cases. The robust model is substantially worse; the permissive and exact-consensus models are better on these averages. No significance test is implied.

Model	Cell F1	Closure exact	Basis F1	Concept F1	Cover F1
Crisp single source	0.723	0.742	0.153	0.138	0.368
Raw majority (2)	0.723	0.742	0.153	0.138	0.368
Source majority (2)	0.723	0.742	0.153	0.138	0.368
WM permissive (1 group)	0.838	0.772	0.266	0.215	0.518
WM robust (2 groups)	0.569	0.704	0.139	0.126	0.250
WM exact consensus	0.830	0.766	0.222	0.204	0.485

Crisp single source at least one positive in the primary layer.

Raw majority (2) at least two positive observation packets.

Source majority (2) at least two nominal positive sources.

WM permissive at least one positive dependence group.

WM robust at least two positive dependence groups.

WM exact consensus at least one positive and no observed negative across the selected layers.

No model was selected or tuned after inspecting outcomes. For every case, CeTTa evaluates the layer projections and checks them against the independent orchestrator, giving 12,150 native semantic assertions.

Point projections are scored against the uncorrupted context by cell F1, held-out closure exactness, concept-set F1, canonical-basis F1, and cover-edge F1. The contextual family is scored by cell and closure coverage together with interval width. Exact duplicate and correlated-copy invariance are separate metrics.

Incomplete-context extension. After the main WM study, but before executing this comparator, we froze a parameter-free lower/upper-completion extension over the same 405 source cases. In the epistemic incomplete-context view, each cell is positive, negative, or unknown, and a completion replaces unknown values by definite ones [5]. We use only the primary evidence layer: the lower completion maps unknown and missing cells to false, and the upper completion maps them to true; both preserve known positive and negative observations. The source layer is reconstructed and must match the original receipt exactly before it is scored. Observed bit flips remain observations in both completions.

Both completion relations are scored separately on the same point and structural metrics. We do not treat their concept families, implication bases, or closures as set-valued intervals: relation inclusion does not in general make those derived families nested. The extension has no fitted parameter, model selection, or post-result tuning and is identified by `cc0596bbeb8e...`

5.2 Point recovery

The raw- and source-majority rows equal the crisp row because the duplicated primary matrix supplies their second positive vote. This equality is not a desirable robustness result: removing the correlated copy changes these projections in almost every case.

Table 5: Frozen incomplete-context comparison on the same 405 cases. Each completion is evaluated as its own crisp relation; no concept-family interval is implied.

Model	Cell F1	Closure exact	Basis F1	Concept F1	Cover F1
Incomplete lower completion	0.723	0.742	0.153	0.138	0.368
Incomplete upper completion	0.736	0.675	0.146	0.050	0.337
WM permissive (1 group)	0.838	0.772	0.266	0.215	0.518

Table 6: WM permissive minus crisp single-source. W/T/L denotes paired cases.

Metric	Mean difference	W/T/L
Cell F1	+0.114	370/1/34
Closure exact	+0.030	168/158/79
Canonical-basis F1	+0.113	212/109/84
Concept F1	+0.076	216/41/148
Cover-edge F1	+0.150	324/18/63

The lower completion is exactly the crisp single-source projection: both retain observed positives and map negative, unknown, and missing cells to non-incidence. A standard incomplete-context construction thus independently recovers the conservative baseline. The WM permissive projection has higher descriptive mean recovery than both completions on every metric in Table 5; the next subsection shows that this point advantage does not dominate the interval comparison.

Table 6 shows paired descriptive differences for the permissive group model. A win/tie/loss is computed per frozen case, so the positive means do not hide that concept and implication recovery sometimes worsens.

5.3 Envelopes and provenance invariance

The three WM incidence projections (permissive, robust, and exact consensus) form the evaluated envelope. Coverage is the fraction of true cell labels or true closure memberships contained in the lower/upper interval. Width is the corresponding average binary width (normalized for closures).

The coverage gain is positive in every frozen missingness and flip-rate stratum, but a wider interval can achieve coverage mechanically. The scientific object is therefore the coverage–width pair, not coverage alone. This experiment does not establish optimal sharpness or calibration of the set-valued interval.

The main protocol also preregistered Brier score and expected calibration error (ECE). For this point-scoring view only, an incidence interval $[l, u]$ is mapped to its midpoint $(l + u)/2 \in \{0, 0.5, 1\}$. The separately frozen completion comparator uses the same map, while the crisp baseline emits hard zero/one forecasts. Table 8 averages the case-level scores; lower Brier and ECE, higher coverage, and lower width are desirable but need not be attained by the same method.

The WM-minus-crisp differences in midpoint Brier score and ECE are negative—the midpoint forecast scores better—in each of the 6 separately frozen missingness and flip-rate strata. This is descriptive evidence that the disagreement state carries useful uncertainty information. It is not an inferential comparison, and midpoint calibration does not imply coverage calibration for the interval. The lower/upper-completion midpoint has lower Brier score and ECE and its interval covers more cells than the WM family, but it is also wider. The fixed comparison therefore supports no claim that either interval uniformly dominates the other.

Table 7: Contextual-envelope coverage and sharpness. Higher coverage is purchased with explicit width; this table is not a point-accuracy comparison.

Target	Crisp coverage	WM coverage	Gain	Mean width
Cells	0.799	0.919	+0.120	0.239
Closures	0.742	0.845	+0.103	0.118

Table 8: Cell-level point forecasts and interval coverage/width. Brier and ECE score interval midpoints; they do not establish calibration of the set-valued intervals.

Forecast	Support	Brier	ECE	Coverage	Width
Crisp hard point	$\{0, 1\}$	0.201	0.201	0.799	0.000
Incomplete completion midpoint	$\{0, 0.5, 1\}$	0.116	0.126	0.968	0.334
WM model-family midpoint	$\{0, 0.5, 1\}$	0.141	0.179	0.919	0.239

Every model is invariant to replaying an evidence layer with the same identity: the observed rate is 1.000. The two dependence-group models are invariant to removing the distinct-source correlated copy at rate 1.000. Raw and source-counting thresholds have rate 0.007. This ablation isolates the benefit of preserving a declared dependence structure: nominal source multiplicity is not allowed to masquerade as independent support.

6 What the WM layer contributes

Conservative exactness. When observations are complete and one extraction model is fixed, WM/FCA reconstructs ordinary FCA exactly. This is supported both by the Lean order isomorphism and by full artifact parity over the exact suite.

Reversible evidence state. Every cell judgment can be traced to active stamps, and forgetting has a precise meaning: recompute from the surviving evidence. The lifecycle tests exercise negative, unknown, missing, conflicting, duplicated, and foreign-context observations rather than only monotone addition. Query-specific closure certificates retain both stages of the Galois derivation: the queried attributes, their extent, and the resulting closure. Implication and concept certificates compose this object with the proposition being checked and its Boolean result. Verification recomputes the two stages from the retained immutable index, while an explanation query exposes active stamped observations for exactly the cells consulted by the derivation. Positive, negative, and tampered certificates are tested. Event-backed states also retain their update history; snapshot states explicitly report that no history was supplied.

Dependence-aware projection. The corruption ablation gives a direct counterexample to treating two nominal sources as two independent observations. Group-count gates remain unchanged under the declared correlated copy; raw and source-counting gates do not. The formal theorem gives the same result for arbitrary finite states satisfying the group-subset premise.

Model-family answers. Robust/permissive relations expose uncertainty that a single matrix suppresses. Here the envelope covers more true cells and closures, with its width reported. The point results also show why the family should not be collapsed by fiat: the robust two-group gate loses badly, while the permissive and exact-consensus gates improve the descriptive averages.

These contributions use the WM calculus rather than a fixed PLN rule set. PLN evidence values can instantiate the evidence carrier and supply additional deduction or revision rules, but none is required for the conservative FCA projection or the provenance-state laws.

7 Limitations

1. The corruption data are synthetic perturbations of controlled-random contexts. They isolate missingness, flips, dependence, and unknowns, but do not establish utility in a natural scientific domain.
2. The analysis is post-run descriptive. The protocol froze models and hypotheses, but did not freeze a sampling model, confidence interval, or significance test. We therefore report no p -values or inferential confidence intervals.
3. Dependence groups are supplied by the modeller. Invariance is exact conditional on the grouping; discovering or validating the groups is outside the current experiment.
4. The envelope comparison does not optimize the coverage-width frontier. It includes a fixed lower/upper incomplete-context comparator, but not a tuned fuzzy, rough, or graded-possibility FCA method. Brier and ECE evaluate only interval midpoints as point forecasts, not calibration of the set-valued intervals.
5. Exact lattice enumeration is restricted by a frozen attribute bound. Wider cases validate exact queries only. We make no general performance claim against specialized FCA engines.
6. Lean proves the abstract semantics and FCA bridge, while runtime equivalence is established by executable differential tests. The C implementation is not extracted from Lean and is not end-to-end verified.
7. The query certificates are replayable state-carrying objects, not minimal cryptographic or independently checkable proof terms. They can expose every consulted cell and its active observations on demand, but do not eagerly materialize those explanations. Snapshot states cannot reconstruct an update history that was not supplied; event-backed states retain one explicitly.

8 Reproducibility

The runtime artifacts live under `benchmarks/wm_fca` in the CeTTa repository [12]. The main protocol is `corruption_protocol.json`; the compressed run receipt and descriptive analysis are under `results`. The staged runtime measurements use `ergonomics_protocol.json` and a separate receipt under `results`. The corruption receipt artifact hash is `835ac74af9be...`, and its decompressed content hash is `40c755197680...`. The incomplete-context extension is identified by `cc0596bbeb8e...`; its receipt artifact and content hashes are `a414d827b9d3...` and `003a0e25831c...`. Exact and query oracles pin source commits, input hashes, dependency versions, matrix semantics, and oracle agreement.

The Lean source is `Mettapedia/Logic/ConceptOntology/WMFCA.lean` in MeTTapedia [13]. Paper tables are emitted by `papers/generate_wm_fca_benchmark_tables.py`. The generator independently replays both the corruption and incomplete-context analyzers from their pinned compressed receipts and requires exact JSON equality with the checked-in analyses. It also checks the

protocol hashes, complete and identical case universes, exact lower-completion/crisp equality, cross-analysis WM means, oracle schemas and agreement, event-prefix cardinality, and every referenced field. For the ergonomics table it checks the runner, library, program, golden-output, and protocol hashes and rederives each reported rate from its median. A missing or drifted field aborts generation. Thus no numeric table entry in this paper is silently defaulted or hand-transcribed.

9 Related work

FCA originates with Wille [1]; the standard mathematical and algorithmic treatment is Ganter and Wille [2]. Fuzzy Galois connections and fuzzy FCA replace Boolean incidence by graded relations [3], while interval-valued FCA represents lower and upper membership degrees [4]. The present construction differs in where uncertainty lives: a provenance state and a family of evidence extraction models precede the crisp relation. Fuzzy or interval evidence can nevertheless be used inside the same interface.

Epistemic incomplete contexts distinguish positive, negative, and unknown incidences and study possible complete contexts obtained by resolving the unknown cells [5]. Our fixed lower/upper-completion comparator instantiates that established boundary on exactly the same corrupted primary layers as the WM models. The WM envelope instead ranges over explicit evidence profiles and extraction gates, so its bounds need not equal the extremal completions of one partial matrix.

The WM formulation is aligned with imprecise-probability practice in retaining a family of admissible precise models instead of prematurely selecting one [6]. Its exact FCA projection and credal concept carriers extend the type-theoretic development in [11]. CeTTa supplies the native MeTTa execution substrate; Lean 4 supplies the proof-checked semantic bridge [10].

10 Conclusion

World-model FCA can retain observation status, provenance, dependence, revision, forgetting, and model uncertainty without sacrificing classical FCA. The crisp theory is recovered exactly as a singleton late projection. The runtime reconstructs independently computed concepts, implications, and lattice covers; wider contexts remain safely query-bounded; incremental updates match fresh recomputation; and the Lean development proves the corresponding abstract laws.

The frozen corruption study provides a first nontrivial use of the additional degrees of freedom. Dependence-aware gates eliminate correlated-copy inflation exactly. Permissive and exact-consensus projections improve several descriptive recovery metrics, while the robust projection worsens them. Contextual envelopes improve coverage with an explicit width cost, and their midpoint point forecasts improve Brier score and ECE in every frozen missingness and flip-rate stratum. Against the fixed incomplete-context baseline, WM permissive has better point and structural recovery, whereas the completion interval has better cell coverage and midpoint calibration at greater width. These results validate the machinery and isolate its behavior, but they do not establish interval calibration or natural-domain superiority. The next scientific step is a preregistered partial-observation dataset in which provenance and dependence are externally meaningful, together with a coverage-sharpness comparison against graded uncertain-FCA methods.

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